

AERIS Expert Review Packet

Anomaly

Source / metric	openaq / PM10 (pm10)
Observed / expected baseline	175 / 28.5455 ug/m3
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-06-10 17:00 UTC
Location	29.7339, -95.2575
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	9 / 638	52.69 to 100; mean 78.2346	58.88 at 2026-06-10 16:55Z; 5 min before event
asos	Air temperature (temperature)	degC	9 / 638	23.3333 to 33.3333; mean 28.7535	31 at 2026-06-10 16:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	9 / 637	0 to 200; mean 140.502	160 at 2026-06-10 16:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	9 / 639	0 to 10.2889; mean 4.5656	6.68777 at 2026-06-10 16:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	10 / 120	5897.03 to 5933.85; mean 5916.48	5920.04 at 2026-06-10 18:00Z; 1 h after event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	10 / 120	134.31 to 1939.63; mean 851.295	1440.93 at 2026-06-10 18:00Z; 1 h after event
noaa_gfs	Precipitable water (precipitable_water)	mm	10 / 120	38.7218 to 52.7703; mean 46.9241	52.6759 at 2026-06-10 18:00Z; 1 h after event
noaa_gfs	Surface pressure (surface_pressure)	hPa	10 / 120	1005.86 to 1016.41; mean 1011.19	1012.08 at 2026-06-10 18:00Z; 1 h after event
noaa_gfs	850-hPa temperature (t_850)	degC	10 / 120	17.0319 to 20.1995; mean 18.6754	17.6074 at 2026-06-10 18:00Z; 1 h after event
noaa_gfs	10-m eastward wind (u_10m)	m/s	10 / 120	-4.7202 to -0.277048; mean -2.2602	-1.98728 at 2026-06-10 18:00Z; 1 h after event
noaa_gfs	10-m northward wind (v_10m)	m/s	10 / 120	0.520322 to 6.25296; mean 3.6685	4.7092 at 2026-06-10 18:00Z; 1 h after event
openaq	Ozone (ozone)	ppm	17 / 1057	0.001 to 0.045; mean 0.0184	0.014 at 2026-06-10 17:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	3 / 182	13 to 175; mean 33.2033	175 at 2026-06-10 17:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	90 / 3781	0 to 39.65; mean 5.0596	23.7 at 2026-06-10 17:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 356	0 to 100; mean 30.4916	37 at 2026-06-10 16:57Z; 2 min before event
openweather	Relative humidity (humidity)	percent	5 / 356	64 to 94; mean 80.3567	71 at 2026-06-10 16:57Z; 2 min before event
openweather	Precipitation rate (precipitation)	mm/h	5 / 356	0 to 0; mean 0	0 at 2026-06-10 16:57Z; 2 min before event
openweather	Surface pressure (pressure)	hPa	5 / 356	1010 to 1017; mean 1013.47	1014 at 2026-06-10 16:57Z; 2 min before event
openweather	Air temperature (temperature)	degC	5 / 356	24.9 to 33.53; mean 29.1841	31.97 at 2026-06-10 16:57Z; 2 min before event
openweather	Wind direction (wind_direction)	degree	5 / 356	0 to 278; mean 146.551	175 at 2026-06-10 16:57Z; 2 min before event
openweather	Wind speed (wind_speed)	m/s	5 / 356	0.45 to 5.94; mean 2.6574	3.58 at 2026-06-10 16:57Z; 2 min before event
purpleair	PM2.5 (pm25)	ug/m3	66 / 4647	0 to 154.9; mean 5.4045	2.5 at 2026-06-10 17:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	9 / 9	1 to 1; mean 1	1 at 2026-06-10 18:32Z; 1 h 33 min after event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-10 18:32Z; 1 h 33 min after event
sentinel5p	CO column (s5p_co_column)	mol/m^2	8 / 8	0.0243048 to 0.026315; mean 0.0254	0.0263147 at 2026-06-10 18:32Z; 1 h 33 min after event
sentinel5p	CO granules available (s5p_co_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-10 18:32Z; 1 h 33 min after event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	8 / 8	7.17019e-05 to 0.000118784; mean 0.0001	0.00010177 at 2026-06-10 18:32Z; 1 h 33 min after event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	9 / 9	1 to 1; mean 1	1 at 2026-06-10 18:32Z; 1 h 33 min after event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	4 / 4	2.25194e-05 to 2.49404e-05; mean 0	2.31696e-05 at 2026-06-10 18:59Z; 2 h after event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-10 18:59Z; 2 h after event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	9 / 9	1 to 1; mean 1	1 at 2026-06-10 18:32Z; 1 h 33 min after event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	8 / 8	-2.60596e-05 to 0.00010131; mean 0	2.51271e-05 at 2026-06-10 18:32Z; 1 h 33 min after event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	9 / 9	1 to 1; mean 1	1 at 2026-06-10 18:32Z; 1 h 33 min after event
tceq	Carbon monoxide (co)	ppm	3 / 217	0.1 to 1; mean 0.206	0.1 at 2026-06-10 17:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	12 / 864	-0.9 to 24.3; mean 3.5725	9 at 2026-06-10 17:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 207	-0.6 to 0.6; mean -0.2193	-0.1 at 2026-06-10 17:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	06-09 00Z 89.54, 06Z 93.03, 12Z 75.61, 18Z 65.65, 06-10 00Z 83.04, 06Z 90.84, 12Z 72.32, 18Z 64.55, 06-11 00Z 82.38, 06Z 89.21, 12Z 73, 18Z 66.87, 06-12 00Z 82.6
asos	Air temperature (temperature)	06-09 00Z 25.78, 06Z 25.53, 12Z 29.44, 18Z 31.23, 06-10 00Z 27.63, 06Z 26.38, 12Z 29.83, 18Z 31.4, 06-11 00Z 27.7, 06Z 26.23, 12Z 30.06, 18Z 31.66, 06-12 00Z 28.02
asos	Wind direction (wind_direction)	06-09 00Z 105.6, 06Z 74.81, 12Z 124.4, 18Z 136.7, 06-10 00Z 141.2, 06Z 143.5, 12Z 157.5, 18Z 155.9, 06-11 00Z 153.4, 06Z 141.7, 12Z 163, 18Z 148.9, 06-12 00Z 151.4
asos	Wind speed (wind_speed)	06-09 00Z 2.058, 06Z 1.484, 12Z 4.554, 18Z 6.069, 06-10 00Z 4.304, 06Z 2.978, 12Z 5.28, 18Z 6.049, 06-11 00Z 4.64, 06Z 3.077, 12Z 5.154, 18Z 6.583, 06-12 00Z 4.893
noaa_gfs	500-hPa geopotential height (gh_500)	06-09 06Z 5930, 12Z 5926, 18Z 5933, 06-10 00Z 5922, 06Z 5928, 12Z 5913, 18Z 5920, 06-11 00Z 5906, 06Z 5912, 12Z 5898, 18Z 5907, 06-12 00Z 5903
noaa_gfs	Planetary boundary layer height (pbl_height)	06-09 06Z 379.7, 12Z 247.3, 18Z 1573, 06-10 00Z 941.7, 06Z 604.8, 12Z 463.8, 18Z 1422, 06-11 00Z 884.1, 06Z 721, 12Z 508.9, 18Z 1546, 06-12 00Z 922.4
noaa_gfs	Precipitable water (precipitable_water)	06-09 06Z 41.71, 12Z 45, 18Z 48.67, 06-10 00Z 47.57, 06Z 46.81, 12Z 48.82, 18Z 51.62, 06-11 00Z 49.74, 06Z 43.94, 12Z 44.99, 18Z 47.17, 06-12 00Z 47.05
noaa_gfs	Surface pressure (surface_pressure)	06-09 06Z 1014, 12Z 1014, 18Z 1014, 06-10 00Z 1012, 06Z 1012, 12Z 1011, 18Z 1011, 06-11 00Z 1009, 06Z 1010, 12Z 1009, 18Z 1009, 06-12 00Z 1008
noaa_gfs	850-hPa temperature (t_850)	06-09 06Z 20.01, 12Z 19.35, 18Z 17.49, 06-10 00Z 18.69, 06Z 18.75, 12Z 18.39, 18Z 17.86, 06-11 00Z 18.65, 06Z 19.11, 12Z 18.31, 18Z 18.3, 06-12 00Z 19.21
noaa_gfs	10-m eastward wind (u_10m)	06-09 06Z -1.991, 12Z -1.546, 18Z -3.591, 06-10 00Z -3.356, 06Z -2.224, 12Z -1.412, 18Z -2.164, 06-11 00Z -2.451, 06Z -1.733, 12Z -1.334, 18Z -2.149, 06-12 00Z -3.172
noaa_gfs	10-m northward wind (v_10m)	06-09 06Z 2.398, 12Z 1.303, 18Z 3.632, 06-10 00Z 3.984, 06Z 3.27, 12Z 2.572, 18Z 5.114, 06-11 00Z 4.572, 06Z 4.551, 12Z 3.235, 18Z 4.892, 06-12 00Z 4.499
openaq	Ozone (ozone)	06-09 00Z 0.02035, 06Z 0.01847, 12Z 0.02151, 18Z 0.02709, 06-10 00Z 0.01891, 06Z 0.01541, 12Z 0.01547, 18Z 0.02051, 06-11 00Z 0.0164, 06Z 0.0141, 12Z 0.01504, 18Z 0.02168, 06-12 00Z 0.01663
openaq	PM10 (pm10)	06-09 00Z 29.67, 06Z 20.08, 12Z 26.7, 18Z 39.33, 06-10 00Z 22.58, 06Z 19.33, 12Z 51.46, 18Z 48.78, 06-11 00Z 21.83, 06Z 24.94, 12Z 44.72, 18Z 44.83, 06-12 00Z 26.11
openaq	PM2.5 (pm25)	06-09 00Z 6.079, 06Z 5.703, 12Z 5.859, 18Z 6.09, 06-10 00Z 5.82, 06Z 5.056, 12Z 5.104, 18Z 4.636, 06-11 00Z 4.811, 06Z 4.361, 12Z 6.3, 18Z 3.559, 06-12 00Z 2.542
openweather	Cloud cover (cloud_cover)	06-09 00Z 18.6, 06Z 9.233, 12Z 32.27, 18Z 58.6, 06-10 00Z 3.2, 06Z 3.3, 12Z 18.46, 18Z 70.57, 06-11 00Z 18.23, 06Z 32.1, 12Z 19.33, 18Z 71.43, 06-12 00Z 29.36
openweather	Relative humidity (humidity)	06-09 00Z 90, 06Z 91.67, 12Z 79.27, 18Z 70.6, 06-10 00Z 82.67, 06Z 90.63, 12Z 78.31, 18Z 69, 06-11 00Z 80.7, 06Z 88.93, 12Z 79.27, 18Z 70.9, 06-12 00Z 80.48
openweather	Precipitation rate (precipitation)	06-09 00Z 0, 06Z 0, 12Z 0, 18Z 0, 06-10 00Z 0, 06Z 0, 12Z 0, 18Z 0, 06-11 00Z 0, 06Z 0, 12Z 0, 18Z 0, 06-12 00Z 0
openweather	Surface pressure (pressure)	06-09 00Z 1016, 06Z 1016, 12Z 1017, 18Z 1015, 06-10 00Z 1015, 06Z 1014, 12Z 1014, 18Z 1012, 06-11 00Z 1012, 06Z 1011, 12Z 1012, 18Z 1011, 06-12 00Z 1012
openweather	Air temperature (temperature)	06-09 00Z 25.98, 06Z 25.8, 12Z 29.83, 18Z 32.03, 06-10 00Z 27.92, 06Z 26.55, 12Z 29.9, 18Z 32.41, 06-11 00Z 28.52, 06Z 26.54, 12Z 29.69, 18Z 32.67, 06-12 00Z 28.92
openweather	Wind direction (wind_direction)	06-09 00Z 156.8, 06Z 127.3, 12Z 131.2, 18Z 141.6, 06-10 00Z 132.1, 06Z 141, 12Z 157, 18Z 166.8, 06-11 00Z 147.4, 06Z 157.2, 12Z 167.2, 18Z 145.3, 06-12 00Z 143.9
openweather	Wind speed (wind_speed)	06-09 00Z 1.678, 06Z 1.962, 12Z 2.326, 18Z 3.86, 06-10 00Z 2.77, 06Z 1.905, 12Z 2.481, 18Z 3.153, 06-11 00Z 2.768, 06Z 2.032, 12Z 2.445, 18Z 3.653, 06-12 00Z 2.678

Source	Metric	Values
purpleair	PM2.5 (pm25)	06-09 00Z 6.539, 06Z 6.46, 12Z 5.565, 18Z 5.708, 06-10 00Z 6.479, 06Z 6.372, 12Z 4.676, 18Z 3.717, 06-11 00Z 5.776, 06Z 7.15, 12Z 6.401, 18Z 3.677, 06-12 00Z 2.766
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	06-09 18Z 1, 06-10 18Z 1, 06-11 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	06-09 18Z 1, 06-10 18Z 1, 06-11 18Z 1
sentinel5p	CO column (s5p_co_column)	06-09 18Z 0.02502, 06-10 18Z 0.02631, 06-11 18Z 0.02512
sentinel5p	CO granules available (s5p_co_granule_available)	06-09 18Z 1, 06-10 18Z 1, 06-11 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	06-09 18Z 0.0001174, 06-10 18Z 0.0001011, 06-11 18Z 8.035e-05
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	06-09 18Z 1, 06-10 18Z 1, 06-11 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	06-09 18Z 2.46e-05, 06-10 18Z 2.317e-05, 06-11 18Z 2.373e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	06-09 18Z 1, 06-10 18Z 1, 06-11 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	06-09 18Z 1, 06-10 18Z 1, 06-11 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	06-09 18Z 1.951e-05, 06-10 18Z 2.942e-05, 06-11 18Z 2.187e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	06-09 18Z 1, 06-10 18Z 1, 06-11 18Z 1
tceq	Carbon monoxide (co)	06-09 00Z 0.1667, 06Z 0.2389, 12Z 0.2529, 18Z 0.2333, 06-10 00Z 0.1778, 06Z 0.1833, 12Z 0.2333, 18Z 0.2235, 06-11 00Z 0.1778, 06Z 0.1889, 12Z 0.2222, 18Z 0.1833, 06-12 00Z 0.1667
tceq	Nitrogen dioxide (no2)	06-09 00Z 4.108, 06Z 5.7, 12Z 5.741, 18Z 3.217, 06-10 00Z 2.989, 06Z 3.764, 12Z 3.993, 18Z 2.963, 06-11 00Z 2.214, 06Z 3.597, 12Z 3.835, 18Z 2.544, 06-12 00Z 2.296
tceq	Sulfur dioxide (so2)	06-09 00Z -0.1667, 06Z -0.2611, 12Z -0.2437, 18Z -0.2294, 06-10 00Z -0.2, 06Z -0.3059, 12Z -0.2647, 18Z -0.1647, 06-11 00Z -0.08333, 06Z -0.2533, 12Z -0.275, 18Z -0.2176, 06-12 00Z -0.1611

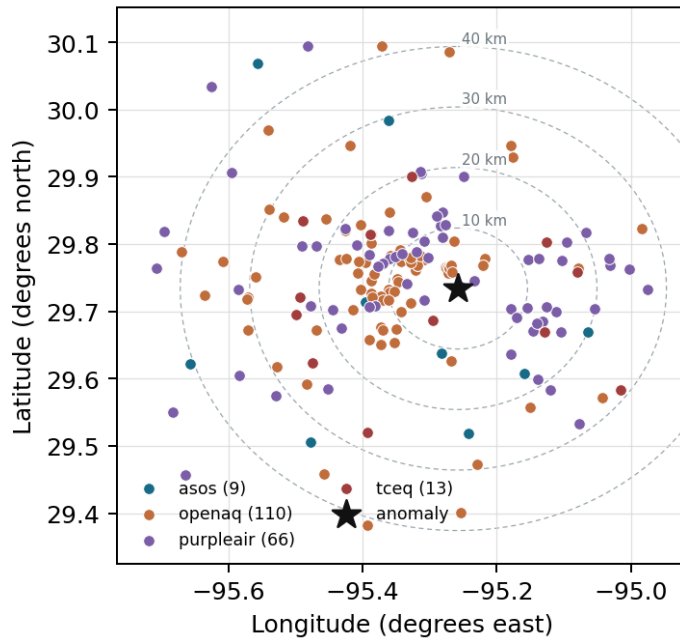
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

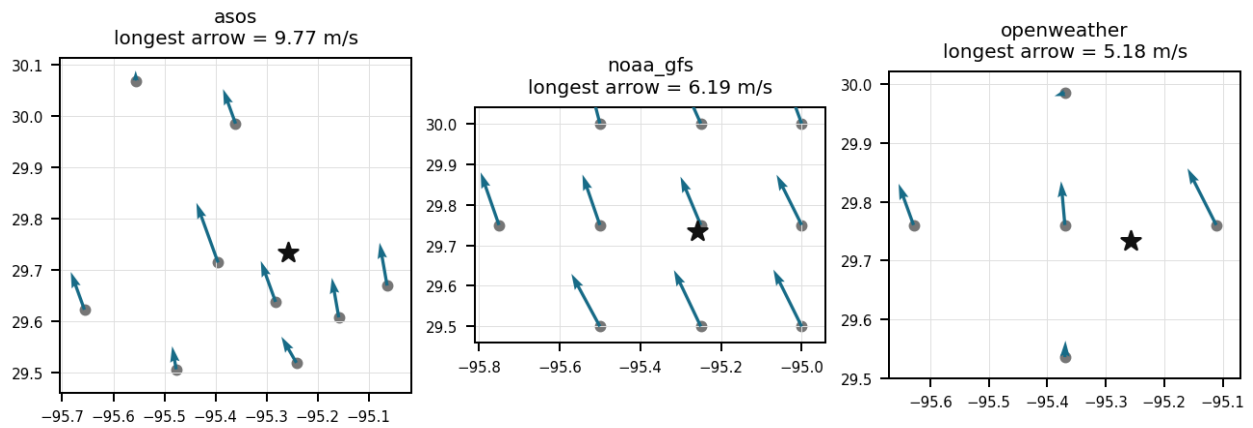
Source	Metric	Values
asos	Relative humidity (humidity)	HOU (10.986 km) 77.09, MCJ (13.461 km) 80.22, EFD (17.008 km) 78.36, T41 (20.017 km) 79.74, LVJ (23.956 km) 76.34, IAH (29.579 km) 78.05, AXH (33.036 km) 76.78, SGR (40.495 km) 80.47, +1 more
asos	Air temperature (temperature)	HOU (10.986 km) 29.04, MCJ (13.461 km) 28.44, EFD (17.008 km) 29.02, T41 (20.017 km) 28.82, LVJ (23.956 km) 28.97, IAH (29.579 km) 28.79, AXH (33.036 km) 28.42, SGR (40.495 km) 28.63, +1 more
asos	Wind direction (wind_direction)	HOU (10.986 km) 148.3, MCJ (13.461 km) 148.3, EFD (17.008 km) 123.5, T41 (20.017 km) 150.8, LVJ (23.956 km) 134.7, IAH (29.579 km) 141.9, AXH (33.036 km) 123.1, SGR (40.495 km) 150, +1 more
asos	Wind speed (wind_speed)	HOU (10.986 km) 5.094, MCJ (13.461 km) 6.352, EFD (17.008 km) 4.769, T41 (20.017 km) 5.673, LVJ (23.956 km) 3.815, IAH (29.579 km) 4.751, AXH (33.036 km) 2.608, SGR (40.495 km) 5.102, +1 more
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.25 (1.931 km) 5917, gfs:29.75,-95.50 (23.481 km) 5917, gfs:29.75,-95.00 (24.925 km) 5917, gfs:29.50,-95.25 (26.019 km) 5916, gfs:30.00,-95.25 (29.598 km) 5917, gfs:29.50,-95.50 (35.014 km) 5916, gfs:29.50,-95.00 (36.001 km) 5916, gfs:30.00,-95.50 (37.713 km) 5917, +2 more

Source	Metric	Values
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.25 (1.931 km) 961, gfs:29.75,-95.50 (23.481 km) 976, gfs:29.75,-95.00 (24.925 km) 803.5, gfs:29.50,-95.25 (26.019 km) 735.6, gfs:30.00,-95.25 (29.598 km) 993.4, gfs:29.50,-95.50 (35.014 km) 727.1, gfs:29.50,-95.00 (36.001 km) 798.7, gfs:30.00,-95.50 (37.713 km) 1062, +2 more
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.25 (1.931 km) 47.08, gfs:29.75,-95.50 (23.481 km) 47.04, gfs:29.75,-95.00 (24.925 km) 46.78, gfs:29.50,-95.25 (26.019 km) 47.51, gfs:30.00,-95.25 (29.598 km) 46.46, gfs:29.50,-95.50 (35.014 km) 47.7, gfs:29.50,-95.00 (36.001 km) 47.25, gfs:30.00,-95.50 (37.713 km) 46.46, +2 more
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.25 (1.931 km) 1012, gfs:29.75,-95.50 (23.481 km) 1011, gfs:29.75,-95.00 (24.925 km) 1013, gfs:29.50,-95.25 (26.019 km) 1012, gfs:30.00,-95.25 (29.598 km) 1010, gfs:29.50,-95.50 (35.014 km) 1011, gfs:29.50,-95.00 (36.001 km) 1013, gfs:30.00,-95.50 (37.713 km) 1009, +2 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.25 (1.931 km) 18.64, gfs:29.75,-95.50 (23.481 km) 18.57, gfs:29.75,-95.00 (24.925 km) 18.78, gfs:29.50,-95.25 (26.019 km) 18.7, gfs:30.00,-95.25 (29.598 km) 18.63, gfs:29.50,-95.50 (35.014 km) 18.7, gfs:29.50,-95.00 (36.001 km) 18.75, gfs:30.00,-95.50 (37.713 km) 18.55, +2 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.25 (1.931 km) -2.039, gfs:29.75,-95.50 (23.481 km) -2.009, gfs:29.75,-95.00 (24.925 km) -2.384, gfs:29.50,-95.25 (26.019 km) -2.498, gfs:30.00,-95.25 (29.598 km) -2.141, gfs:29.50,-95.50 (35.014 km) -2.73, gfs:29.50,-95.00 (36.001 km) -2.873, gfs:30.00,-95.50 (37.713 km) -1.712, +2 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.25 (1.931 km) 3.444, gfs:29.75,-95.50 (23.481 km) 3.543, gfs:29.75,-95.00 (24.925 km) 3.915, gfs:29.50,-95.25 (26.019 km) 3.676, gfs:30.00,-95.25 (29.598 km) 3.32, gfs:29.50,-95.50 (35.014 km) 3.369, gfs:29.50,-95.00 (36.001 km) 4.428, gfs:30.00,-95.50 (37.713 km) 3.543, +2 more
openaq	Ozone (ozone)	3378 (0.0 km) 0.01522, 2039 (5.211 km) 0.01812, 325 (6.39 km) 0.01868, 2046 (10.787 km) 0.01568, 320 (12.079 km) 0.01394, 1319333 (14.037 km) 0.02116, 332 (14.341 km) 0.02311, 3214 (14.841 km) 0.01714, +9 more
openaq	PM10 (pm10)	13380082 (0.0 km) 45.18, 15852419 (7.015 km) 37.76, 271 (14.341 km) 19.18
openaq	PM2.5 (pm25)	3379 (0.0 km) 8.992, 15539682 (0.019 km) 6.132, 8967123 (0.061 km) 6.598, 8967479 (0.061 km) 6.145, 9489522 (2.875 km) 5.39, 8967065 (2.921 km) 5.384, 9201801 (3.61 km) 4.864, 12178865 (3.7 km) 5.269, +82 more
openweather	Cloud cover (cloud_cover)	grid:center (11.235 km) 29.03, grid:east (14.438 km) 27.7, grid:south (24.557 km) 27.28, grid:north (29.948 km) 37.61, grid:west (35.938 km) 30.86
openweather	Relative humidity (humidity)	grid:center (11.235 km) 80.26, grid:east (14.438 km) 83.49, grid:south (24.557 km) 78.99, grid:north (29.948 km) 79.69, grid:west (35.938 km) 79.35
openweather	Precipitation rate (precipitation)	grid:center (11.235 km) 0, grid:east (14.438 km) 0, grid:south (24.557 km) 0, grid:north (29.948 km) 0, grid:west (35.938 km) 0
openweather	Surface pressure (pressure)	grid:center (11.235 km) 1013, grid:east (14.438 km) 1014, grid:south (24.557 km) 1013, grid:north (29.948 km) 1013, grid:west (35.938 km) 1013
openweather	Air temperature (temperature)	grid:center (11.235 km) 29.41, grid:east (14.438 km) 29.07, grid:south (24.557 km) 29.01, grid:north (29.948 km) 29.24, grid:west (35.938 km) 29.19
openweather	Wind direction (wind_direction)	grid:center (11.235 km) 145.5, grid:east (14.438 km) 148.4, grid:south (24.557 km) 163.3, grid:north (29.948 km) 122, grid:west (35.938 km) 153.6
openweather	Wind speed (wind_speed)	grid:center (11.235 km) 2.677, grid:east (14.438 km) 4.298, grid:south (24.557 km) 1.988, grid:north (29.948 km) 2.469, grid:west (35.938 km) 1.854
purpleair	PM2.5 (pm25)	99797 (0.58 km) 4.837, 166435 (2.659 km) 6.692, 6752 (5.292 km) 6.693, 222601 (6.69 km) 1.974, 133994 (7.384 km) 0.04521, 235425 (8.236 km) 5.44, 229853 (8.533 km) 1.908, 127441 (8.715 km) 6.679, +58 more
tceq	Carbon monoxide (co)	482011035 (0.021 km) 0.1233, 482011039 (14.356 km) 0.1352, 482011052 (15.437 km) 0.3575
tceq	Nitrogen dioxide (no2)	482011035 (0.021 km) 5.446, 482010416 (6.399 km) 2.73, 482011039 (14.356 km) 2.519, 482010026 (14.86 km) 7.258, 482011052 (15.437 km) 8.5, 482011015 (17.424 km) 2.668, 482010024 (19.728 km) 2.564, 482011066 (22.748 km) 2.951, +4 more
tceq	Sulfur dioxide (so2)	482011035 (0.021 km) -0.1583, 482011039 (14.356 km) 0.02258, 482010051 (24.254 km) -0.4849

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide

Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.

Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.

Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.

Missing or thin evidence means Unsure, not Invalid.

A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.

Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 35

The air quality anomaly in Houston, Texas is likely caused by a combination of factors including high levels of PM2.5 and NO2 emissions from industrial activities and vehicle traffic.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 2 of 35

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 3 of 35

The air quality anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter and volatile organic compounds.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 4 of 35

Industrial bulk material handling or fugitive dust emissions contributed to the PM10 spike at openaq without accompanying increases in combustion-derived gaseous pollutants measured by tceq.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 5 of 35

Localized mechanical dust resuspension from surface disturbance or nearby construction drove the coarse particulate spike recorded by openaq while ambient fine particle levels remained low on purpleair.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 6 of 35

The anomaly occurred on June 10th, 2026, at approximately 17:00:00+00:00.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 7 of 35

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

Mark exactly one:

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Note (optional):

Claim 8 of 35

Meteorological conditions in Houston at the anomaly time were favorable for transport of mechanically generated dust because surface winds were southerly to south-southeasterly at about 3.6 to 6.7 m/s and afternoon boundary-layer height was deep near 1441 m.

Mark exactly one:

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Note (optional):

Claim 9 of 35

An OpenAQ monitoring station in Houston registered a severe PM10 concentration spike to 175.0 $\mu\text{g}/\text{m}^3$ on June 10, 2026, at 17:00 UTC.

Mark exactly one:

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Note (optional):

Claim 10 of 35

The PM10 anomaly value of 175.0 ug/m3 exceeded the expected value of 28.545454545454547 ug/m3 and had a z-score of 9.414583354484428, which is labeled severe in the anomaly record.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 11 of 35

At the time of the PM10 anomaly on 2026-06-10T17:00:00+00:00, nearby PM2.5 concentrations measured 23.7 µg/m³ on openaq and 2.5 µg/m³ on purpleair, indicating that the spike was predominantly coarse particulate matter.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 12 of 35

The air quality anomaly in Houston, Texas is not caused by changes in temperature or wind direction.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 13 of 35

Regional combustion smoke or widespread secondary aerosol production is contradicted because ozone was low at 0.014 ppm, TCEQ carbon monoxide was only 0.1 ppm, TCEQ sulfur dioxide was -0.1 ppb, and satellite carbon monoxide and nitrogen dioxide columns were not exceptionally elevated at the overpass time.

Mark exactly one:

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Note (optional):

Claim 14 of 35

The nearest OpenAQ PM2.5 measurement at the anomaly location and time was 23.7 ug/m3, while the nearest PurpleAir PM2.5 measurement 0.58 km away at 2026-06-10T17:00:00+00:00 was 2.5 ug/m3.

Mark exactly one:

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Note (optional):

Claim 15 of 35

Nearby PurpleAir and OpenAQ sensors maintained low fine particulate matter (PM2.5) averages below 10 µg/m³ during the PM10 spike, isolating the anomaly to coarse particulate matter.

Mark exactly one:

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Note (optional):

Claim 16 of 35

The isolated PM10 concentration spike recorded by openaq supports a localized source dominated by coarse-mode particles such as fugitive dust or construction activity.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 17 of 35

Surface meteorological observations near the anomaly recorded warm temperatures around 31 °C to 32 °C, relative humidity below 60%, and no precipitation, providing dry atmospheric conditions conducive to mechanical dust resuspension.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 18 of 35

Surface meteorology recorded by asos and openweather showed dry conditions, warm temperatures, and moderate wind speeds that facilitated local dust resuspension.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 19 of 35

The temperature and wind speed data suggest that the air quality anomaly was influenced by a heatwave event with temperatures reaching up to 33.53°C (92.5°F) and wind speeds of up to 3.58 m/s.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 20 of 35

A nearby dust source such as road dust, construction activity, or industrial material handling is meteorologically plausible because surface winds near the anomaly were southerly to south-southeasterly at about 6.7 m/s, no rain was reported, and the daytime boundary layer was deep near 1441 m, which supports transport of freshly generated coarse particles from upwind local sources rather than stagnation.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 21 of 35

Moderate surface wind speeds and strong daytime boundary layer convective mixing captured by asos and gfs enhanced local dust resuspension and transport across the area.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 22 of 35

The PM2.5 levels are elevated at 4.837 ug/m3, which is higher than the mean of 5.4045 ug/m3.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 23 of 35

The anomaly is due to a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, which have led to the accumulation of pollutants in the Houston area.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 24 of 35

Local dust or soil entrainment is a leading physical cause of the PM10 spike because PM10 reached 175 ug/m3 while nearby PM2.5 remained much lower at 23.7 ug/m3 and nearby PurpleAir PM2.5 was only 2.5 ug/m3, a size pattern that is more consistent with coarse particles than with a regional fine-combustion plume.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 25 of 35

Low PM2.5 concentrations measured across the surrounding network by openaq and purpleair contradict a regional combustion or smoke transport event during the PM10 anomaly.

Mark exactly one:

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Note (optional):

Claim 26 of 35

The observed air-quality anomaly is a PM10 anomaly with a measured concentration of 175.0 ug/m3 at 2026-06-10T17:00:00+00:00 near 29.734, -95.257 in Houston.

Mark exactly one:

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Note (optional):

Claim 27 of 35

An openaq monitoring sensor located at coordinates (29.734, -95.257) recorded a severe PM10 concentration anomaly of 175.0 $\mu\text{g}/\text{m}^3$ at 2026-06-10T17:00:00+00:00, significantly exceeding the expected baseline of 28.55 $\mu\text{g}/\text{m}^3$ with a z-score of 9.41.

Mark exactly one:

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Note (optional):

Claim 28 of 35

Mechanical resuspension of road dust or industrial yard dust is a leading physical cause of the PM10 spike because surface winds near the anomaly were from the south-southeast at roughly 6.7 m/s and the daytime boundary layer was deep, conditions that favor lifting and transport of coarse particles from nearby surfaces and facilities.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 29 of 35

Regional combustion or an industrial gas-rich plume is not strongly supported at the anomaly time because local CO was 0.1 ppm, local SO₂ was -0.1 ppb, local ozone was 0.014 ppm, and satellite CO and NO₂ columns were not unusually elevated relative to adjacent days.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 30 of 35

The air quality anomaly in Houston, Texas is not caused by high levels of CO, NO₂, or SO₂ emissions.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 31 of 35

Construction activity or bulk-material handling near the monitor is a plausible physical cause of the PM10 spike because the event was highly localized in PM10 and was not accompanied by strong co-pollutant signals, with low local CO, modest NO2, and near-zero SO2 arguing against a dominant fresh combustion or sulfur-rich industrial plume.

Mark exactly one:

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Note (optional):

Claim 32 of 35

Local mechanical resuspension or dust emission is supported because the PM10 concentration reached 175 ug/m3 while nearby PM2.5 remained low to moderate at 23.7 ug/m3 in openaq and 2.5 ug/m3 in purpleair, a pattern characteristic of coarse particles rather than fine-particle regional haze.

Mark exactly one:

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Note (optional):

Claim 33 of 35

The CO, SO2, and O3 levels were also elevated during the air quality anomaly, indicating a potential contribution from industrial or vehicular sources.

Mark exactly one:

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Note (optional):

Claim 34 of 35

The Houston PM10 anomaly is most consistent with a local coarse-particle dust plume rather than a regional fine-particle pollution episode because PM10 reached 175 ug/m3 while nearby PM2.5 remained much lower at 23.7 ug/m3 in openaq and 2.5 ug/m3 in purpleair at the same time.

Mark exactly one:

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Note (optional):

Claim 35 of 35

A severe thunderstorm or tornado event in the Houston area has stirred up dust and debris, contributing to the air quality anomaly.

Mark exactly one:

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Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.

